Control Systems

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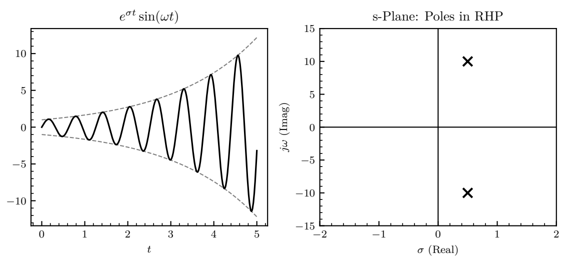
Laplace Transform

Definition



The Laplace transform is a generalized Fourier transform used to simplify differential equations, either transforming PDEs into ODEs or converting ODEs into algebraic expressions. It is specifically designed to handle functions that grow exponentially at infinity. By definition, we have:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty f(t)e^{-st}, dt$$

In control theory, this transform is used to convert the time-domain equations of a dynamic system into a s-domain transfer function, making it easy to see if the natural response is unbounded (i.e., identifying poles with positive real parts).

Let $f(t) = e^{\sigma t} sin(\omega t), t \geq 0$, be a modulated signal, where σ is the growth rate and ω is the angular frequency. Taking the Laplace transform yields

$$F(s) = \frac{\omega}{(s-\sigma)^2 + \omega^2},$$

by setting the denominator to zero

$$\boxed{s = \sigma \pm j\omega = 0.5 \pm j10},$$

because $\mathrm{Re}(s) = \sigma > 0$ these poles lie in the Right-Half Plane (RHP), representing a unstable, exponentially growing oscillation of the time-domain signal.

Essential pairs

$f(t), t > 0$	F(s)
$\delta(t)$	1
u(t) (unit step)	$\frac{1}{s}$
t (unit ramp)	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
e^{at}	$\frac{1}{s+a}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

Properties

These equations represent the differentiation and integration properties of the Laplace transform,

$$\mathcal{L}\{\dot{f}\} = sF(s) - f(0),$$
$$\mathcal{L}\{\ddot{f}\} = s^2F(s) - sf(0) - \dot{f}(0),$$
$$\mathcal{L}\left\{\int_0^t f d\tau\right\} = \frac{F(s)}{s},$$

But in general we almost always assume zero initial conditions so $f(0) = 0$.

The final value theorem is valid if and only if all poles of $sF(s)$ are in the Left-Half Plane (LHP),

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s),$$

And is a tool used to verify a steady-state output or final value as time approaches infinity.

From frequency domain to time domain

Let $F(s)$ be a rational Laplace transform from which we wish to find the time response $f(t)$. The most direct approach is to expand $F(s)$ into a sum of simpler terms.
For distinct real poles, the expansion is given by:

$$F(s) = \frac{N(s)}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b},$$

$$A = \lim_{s\to -a}(s+a)F(s), \quad B = \lim_{s\to -b}(s+b)F(s).$$

Finding A and B and then applying the inverse Laplace transform is straightforward.

However, if $F(s)$ contains a second-order repeated pole at $s = -a$, the function takes the form:

$$F(s) = \frac{N(s)}{(s+a)^2} = \frac{A_1}{s+a} + \frac{A_2}{(s+a)^2},$$

$$A_2 = \lim_{s\to -a}(s+a)^2F(s), \quad A_1 = \lim_{s\to -a}\frac{d}{ds}\left[(s+a)^2F(s)\right].$$

The derivative is required for A_1 to isolate the lower-power coefficient. The resulting inverse Laplace transform introduces a time-multiplied exponential mode:

$$\boxed{f(t) = A_1e^{-at} + A_2te^{-at}, \quad t \geq 0}$$

Transfer Functions and Block Algebra

Derivation

The input (u) and output (y) behavior of a linear time-invariant (LTI) system can be modeled by a linear differential equation of the form:

$$a_0\frac{d^ny}{dt^n} + a_1\frac{d^{n-1}y}{dt^{n-1}} + \cdots + a_ny = b_0\frac{d^mu}{dt^m} + b_1\frac{d^{m-1}u}{dt^{m-1}} + \cdots + b_mu,$$

where $n \geq m$ and $a_k, b_k \in \mathbf{R}$. To derive the transfer function $G(s)$, we apply the Laplace transform term by term assuming zero initial conditions ($u(t) \to U(s)$, $y(t) \to Y(s)$) and solve for the ratio of output to input:

$$G(s) = \frac{Y(s)}{U(s)} = \left.\frac{\mathcal{L}\{\text{output}\}}{\mathcal{L}\{\text{input}\}}\right|_{\text{zero initial conditions}}$$

This function is an inherent property of the system dynamics alone and does not provide information regarding its physical structure.

Transfer Matrix

For multiple-input, multiple-output (MIMO) systems featuring r inputs and m outputs, the system properties are represented by the transfer matrix $\mathbf{G}(s) \in \mathbf{R}^{m\times r}$. Given the output vector $\mathbf{y}$ and input vector $\mathbf{u}$:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_r \end{bmatrix}$$

The transfer matrix can be directly obtained from the state-space representation as:

$$\mathbf{G}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$

where $\mathbf{A} \in \mathbf{R}^{n\times n}$ is the state (dynamics) matrix, $\mathbf{B} \in \mathbf{R}^{n\times r}$ is the input (control) matrix, $\mathbf{C} \in \mathbf{R}^{m\times n}$ is the output (sensor) matrix, and $\mathbf{D} \in \mathbf{R}^{m\times r}$ is the direct transmission (feedforward) matrix, the latter matrix being non-zero if we are modelling a static system or when we have a large sampling interval.

Block Algebra

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